

Question 3**Probability of Event A**

$$P(A) = \frac{n(A)}{N}$$

(a)

$$\text{i. } P(L) = \frac{120}{200} = 0.6000$$

$$\text{ii. } P(T) = \frac{80}{200} = 0.4000$$

$$\text{iii. } P(L \cap T) = \frac{50}{200} = 0.2500$$

General Addition Rule

$$P(L \cup T) = P(L) + P(T) - P(L \cap T)$$

(b)

$$P(L \cup T) = 0.6000 + 0.4000 - 0.2500 = 0.7500$$

Conditional Probability

$$P(T|L) = \frac{P(T \cap L)}{P(L)}$$

(c)

$$P(T|L) = \frac{0.2500}{0.6000} \approx 0.4167$$

Independence Check

$$P(L) \times P(T) = P(L \cap T)$$

(d)

Calculation: $0.6000 \times 0.4000 = 0.2400$.

No, the events are not independent because $0.2400 \neq 0.2500$.

Multiplication Rule (Without Replacement)

$$P(L_1 \cap L_2 \cap L_3) = P(L_1) \times P(L_2|L_1) \times P(L_3|L_1 \cap L_2)$$

(e)

$$P(L_1 \cap L_2 \cap L_3) = \left(\frac{120}{200}\right) \times \left(\frac{119}{199}\right) \times \left(\frac{118}{198}\right) \approx 0.2138$$